

PRT565 MACHINE LEARNING , ARTIFICIAL INTELLIGENCE AND ALGORITHMS

Assessment 03

Value: 30%

Submission Deadline: *27th September 2026; 22:00 (Sydney Time)*

Submission Format

Online research is allowed for all the tasks (you need to add References in APA format).

In case if you have received an extension, please add the screenshot at the back of your assessment file or before starting the tasks.

Rubrics: Please see the last page.

Deductions:

- Delay penalty - 10% for each day without an approval.
- Failing to provide a cover page or submitting with an incomplete cover page will attract a deduction of 15%.
- If online resources have been used, the Referencing is must and failure to provide valid references will attract a deduction of 15%.

Once the marks for this assessment is released, a relevant Announcement will be sent out, usually in 3 weeks' time from submission deadline. If you have any doubt regarding the marking, please get in touch with the Lecturer **within 5 business days** from the date of outcome release for this assessment, after which, your submission will not be re-evaluated for any issues that you may raise. In case you have decided not to submit the assessment at all within the deadline, and want to notify or discuss your options, you Must get in touch with your Lecturer on or before the submission deadline, otherwise no request for submission will be entertained after the submission deadline.

Assessment title	Volume	Value (of total mark)	Related outcomes
Artificial Intelligence Group presentation	6-8 minutes presentation each	30%	1,2,3,4,5,6,7

Group Presentation

Group Formation: A maximum of 3 Members in each group.

Aim: Develop AI based models using general Machine Learning (use Decision Tree, Random Forest and Naïve Bayes, plus any other algorithms as you see fit) and Deep Learning (Multilayer ANNs). Now compare the results of these three models through various metrics such as Accuracy, Precision, Recall, F1-Score etc.

Note: You are not limited to four models, to make the study comprehensive, you can add other models using other Deep Learning Algorithms such as LSTM or RNN.

Development Platform: Python (using Jupyter Notebook or any other IDE of your choice).

Dataset: Use any relevant datasets from <https://www.kaggle.com/datasets> OR <https://archive.ics.uci.edu/ml/datasets.php>. If you want to use any other dataset please email the Unit Coordinator first.

Deliverables: Each of the members must present parts of the overall presentation (maximum of 15 minutes for the full group, 6 minutes max for each member). The delivered report will have the necessary cover page as the first page mentioning the Unit Information, Group Number, Member Information and Campus. The second page should have the link to the recorded presentation along with Project title, a brief description of the project and motivation for selecting the topic. The presentation file will also have to be submitted in pptx format. You can Zip both the Report and the Presentation file before the submission. All the members should submit this same Zip file in Learnline under their group, you can host your group presentation video on YouTube, Google Drive, One Drive or any other file hosting services. However, ensure that the file is open to public and anyone with your Link can View/Download the file. Test it before submitting.

Note: The presentation can have a number of useful points such as (but not limited to):

- Group Information and Project Title
- Introduction\Motivation
- Dataset Description
- Dataset Analysis, Feature Selection, Transformation/Preprocessing & Engineering
- Algorithm and Model Overview including Testing
- Model Performance Evaluation (Testing)
- Conclusion

It is critically important to be loud and clear while doing the presentation, you must be confident and well-organized (Points will be awarded on the quality of individual presentation).

Examples: The presentation examples below can be useful

- ✓ <https://www.youtube.com/watch?v=yIGaReLrTe8>
- ✓ <https://www.youtube.com/watch?v=rw-7V-2oFjg>
- ✓ <https://www.youtube.com/watch?v=wYXtN-5SBcw>

Assessment 3 - Presentation Rubric	
Segments	Max Marks
Preparedness	6.5
*Structure	6.5
*Depth	7
Delivery (Tone, Volume & Pace)	10
	30
*Applicable at group level	
